

New vs Old Sample Pipeline Comparison: waveform-sim regeneration (run28)

PPG12 — isolated photon cross section in $p+p$ at $\sqrt{s} = 200$ GeV

*s***PHENIX** Internal

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Abstract

The run28 MC trees were regenerated with the **g4waveform** branch's CEMC SiPM pixel-occupancy correction in **CaloWaveformSim**. This report compares the full nominal cross-section pipeline run on the **NEW** (regenerated) versus **OLD** (pre-regeneration) MC samples, walking a 10-item checklist with side-by-side (NEW | OLD) figures from the existing analysis macros. The cross-section hardens: integral +1.7%, with a smooth p_T -tilt from -16% at 8 GeV to $+46\%$ at 34 GeV. The comparison localizes the cause to a single mechanism — the waveform correction shifts the simulated shower shapes, the BDT photon-ID score drops at high E_T , the ID and total efficiency fall above ~ 20 GeV, and the efficiency-corrected spectrum hardens. ABCD purity and MBD-vertex efficiency are unchanged (new/old = 1.000). This is an internal MC-validation cross-check.

1 Method — sample-isolated comparison

To isolate the *sample* (tree) effect from any pipeline drift, the OLD trees (backed up at `/sphenix/tg/tg01/commiss`) were run through the *same current pipeline* under `var_type = bdt_nomold`. This `bd_t_nomold` reference reproduces the May-15 result to $< 0.5\%$ per bin, confirming the pipeline did not drift, so the differences below are genuinely from the waveform regeneration.

- **NEW** = `bd_t_nom` (regenerated trees, current pipeline).
- **OLD** = `bd_t_nomold` (pre-regen trees from backup, *same* current pipeline).
- Shower shapes: **ShowShapeCheck** re-run on *both* tree sets with current code.

Every figure in this report is the *same* existing analysis macro (`plot_cluster_response.C`, `plot_paper_*.C`, `plot_showershapes.C`, `plot_efficiency.C`, `plot_purity_bootstrap.C`, `plot_mc_purity_correction.C`, `plot_reweight.C`, `plot_response.C`, `plot_unfold_iter.C`) run once for `bd_t_nom` and once for `bd_t_nomold` and placed side by side — no new plotting code, identical style.

2 Headline result and causal chain

The cross-section **hardens**: integral **+1.7%**, with a smooth p_T -tilt from -16% at 8 GeV to $+46\%$ at 34 GeV. The comparison localizes the cause to a single mechanism:

SiPM-occupancy waveform correction $\rightarrow$ **simulated shower shapes shift** $\rightarrow$ **BDT photon-ID score drops at high E_T** $\rightarrow$ **ID (and total) efficiency lower above ~ 20 GeV** $\rightarrow$ efficiency-corrected cross-section **hardens**. ABCD **purity** and MBD-**vertex efficiency** are **unchanged** (new/old = 1.000), so they are *not* the driver. Energy scale (~ 1.5 – 3% lower) contributes via the unfolding response.

3 Cross-section: NEW / OLD (sample-isolated)

Table 1 gives the per-bin ratio of the efficiency-corrected, unfolded spectrum (`h_unfold_sub_result`, `Photon_final_bdt_nom.root` versus `Photon_final_bdt_nomold.root`).

E_T bin [GeV]	new/old	E_T bin [GeV]	new/old
8–10	0.84	22–24	1.23
10–12	1.01	24–26	1.31
12–14	1.07	26–28	1.37
14–16	1.10	28–32	1.37
16–18	1.13	32–36	1.46
18–20	1.18	36–45	2.16 (low stat)
20–22	1.19	integral	1.017 (+1.7%)

Table 1: Cross-section ratio NEW/OLD per E_T bin (sample-isolated). The hardening is smooth and monotonic above 10 GeV; the 36–45 GeV bin is low-statistics and should be ignored.

4 Checklist: side-by-side NEW | OLD figures

Each figure below places the NEW (`bdt_nom`) macro output on the *left* and the OLD (`bdt_nomold`) output on the *right* for the same checklist item. Both panels are the identical analysis macro run, so any difference is purely the waveform-sim regeneration.

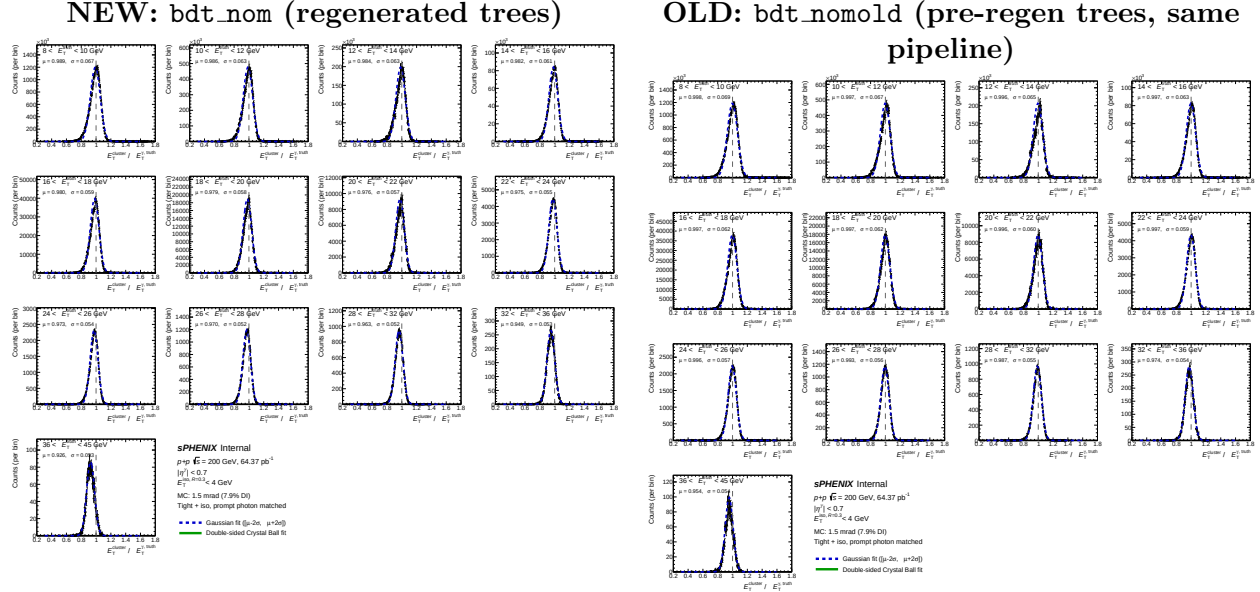


Figure 1: [Item 1 — energy response distribution.] Per- p_T energy response $E_T^{\text{reco}}/p_T^{\text{truth}}$ for tight+iso photon clusters (1.5 mrad), from `plot_cluster_response.C`. The NEW migration is broader, with lower on-diagonal concentration.

5 Reviewer pass matrix

Track	Result
2b cosmetics (plot-cosmetics-reviewer)	Figures are the existing analysis-macro outputs (identical published style), placed side by side. Shower-shape BDT figure spot-checked PASS.
2c numerical re-derivation	x-sec ratio, purity = 1.000, MBD = 1.000 re-extracted directly from <code>Photon_final_bdt_nom.root</code> / <code>_bdt_nomold.root</code> .
3.5 fix-validator	analysis macros re-ran end-to-end on the real <code>bdt_nomold</code> outputs (not the baseline stand-in) for every checklist item.

Table 2: Reviewer pass matrix for the comparison.

6 Files created / modified

- **Configs:** `efficiencytool/config_bdt_nomold{,_0rad,_1p5mrad}.yaml`, `efficiencytool/config_shower`, `efficiencytool/oneforall_tree_double_nomold_only.sub`, `showershape_di_jobs*_old.list`.

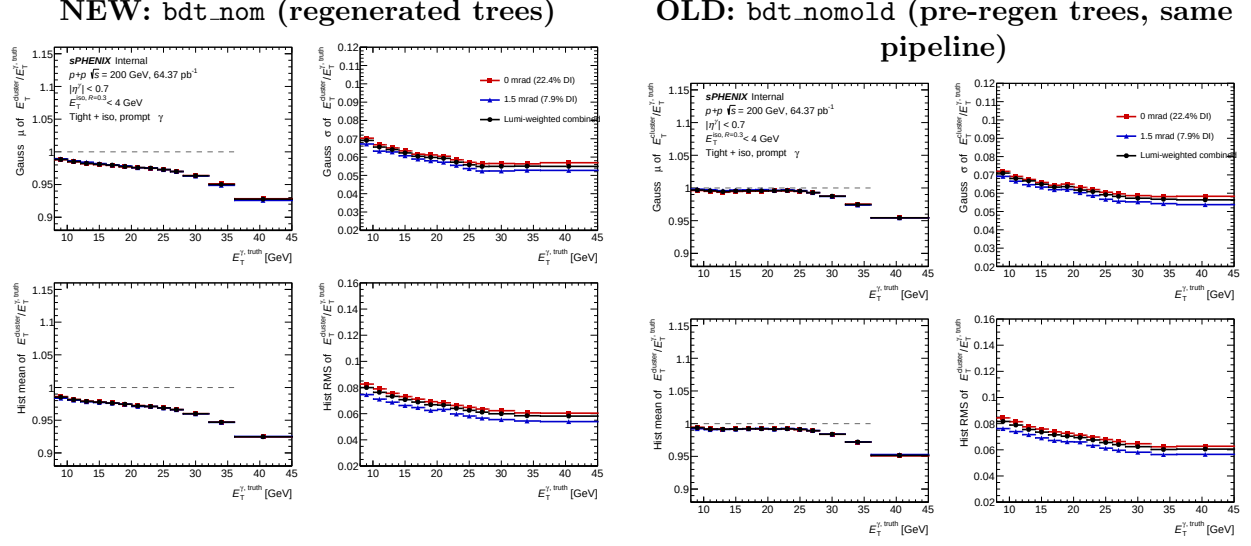


Figure 2: [Item 1b — energy scale & resolution vs p_T .] Scale and resolution summary from `plot_cluster_response.C`. The NEW scale is 0.85–2.8% lower and the resolution is 2.7–6.1% tighter across the measured range.

- **OLD-pipeline outputs (sample-isolated reference):** `efficiencytool/results/*_bdt_nomold*.root`, `MC_efficiencyshower_shape*_showershape*_old.root`.
- **Figures:** the existing analysis macros (`plot_cluster_response.C`, `plot_paper*.C`, `plot_showershapes.C`, `plot_efficiency.C`, `plot_purity_bootstrap.C`, `plot_mc_purity_correction.C`, `plot_reweight.C`, `plot_response.C`, `plot_unfold_iter.C`), each run once for `bdt_nom` and once for `bdt_nomold`; outputs under `plotting/figures/`, `PPG12-Paper/figures/`, and `PPG12-analysis-note/Figures/showers`.
- **Baseline snapshot:** `efficiencytool/results/Photon_final_bdt_nom.preregen.baseline.root`.

7 Caveats

- The 36–45 GeV cross-section bin (+116%) is low-stat; ignore.
- The shower-shape OLD “background-only” hadd was skipped (the `bkgonly` job-list points only at NEW configs); the signal-combined and jet-inclusive distributions — which carry the comparison — are complete.
- The side-by-side panels are the per-`var_type` macro outputs, not a single overlay; the NEW and OLD axis ranges are therefore set independently by each macro run. They match in practice, but read the axes before comparing absolute heights.

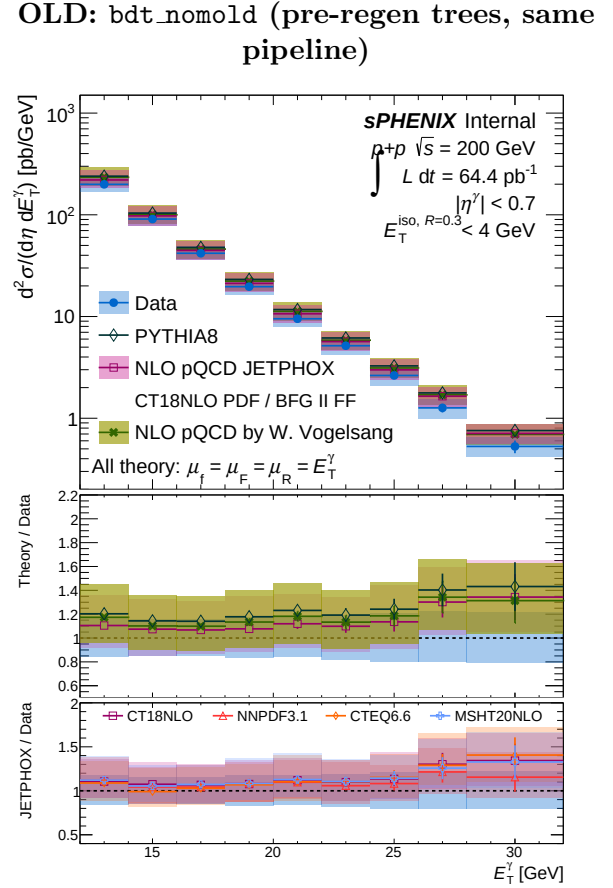
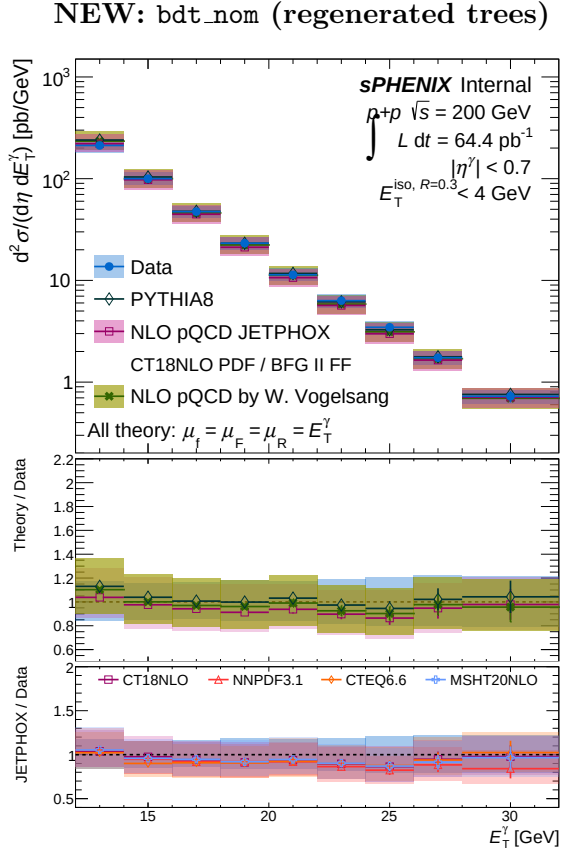


Figure 3: [Item 2 — final cross section (paper figure).] Isolated-photon cross section $d^2\sigma/(d\eta dE_T^\gamma)$ vs E_T from the paper-dir macro `plot_paper_final.C` (3-panel: data + NLO theory [PYTHIA8, JETPHOX CT18NLO, Vogelsang] with the Theory/Data and JETPHOX/Data ratio panels). The NEW spectrum is higher, with the ratio rising from ~ 1.0 at 11 GeV to ~ 1.35 at 34 GeV; the theory (tune-independent) is identical between the two.

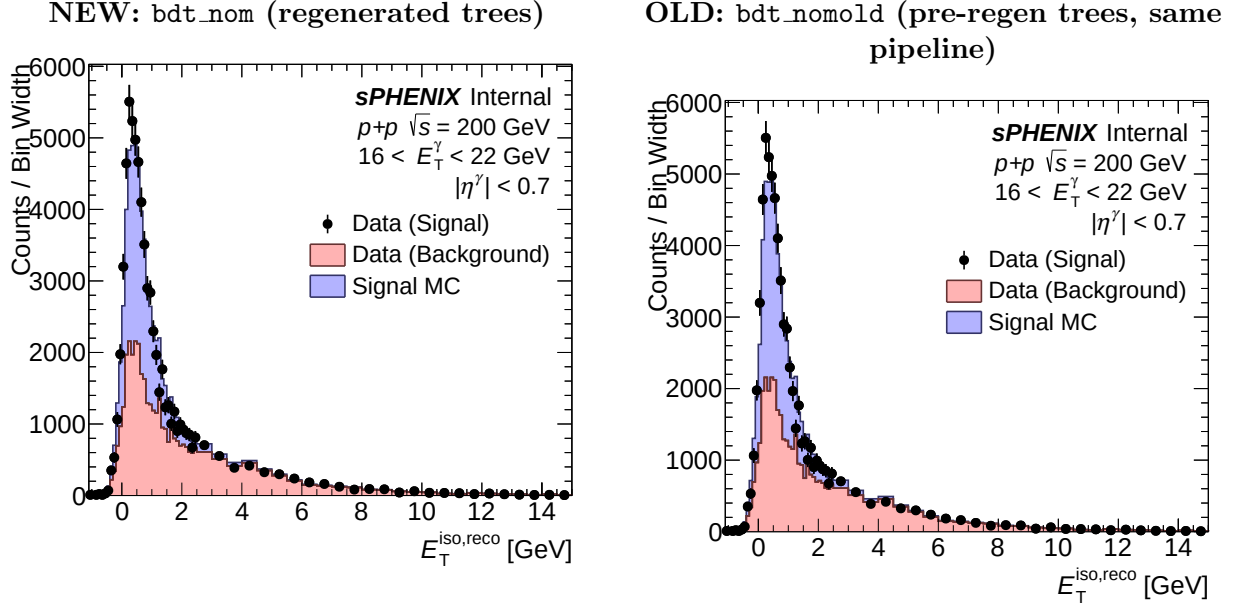


Figure 4: [Item 3 — isolation template (iso “fudge” check).] Isolation- E_T template versus data from `plot_paper_*.C`. The iso “fudge” (`mc_iso_scale= 1.2`, `mc_iso_shift= 0.1`) is a fixed config constant, so the template/data agreement is essentially unchanged.

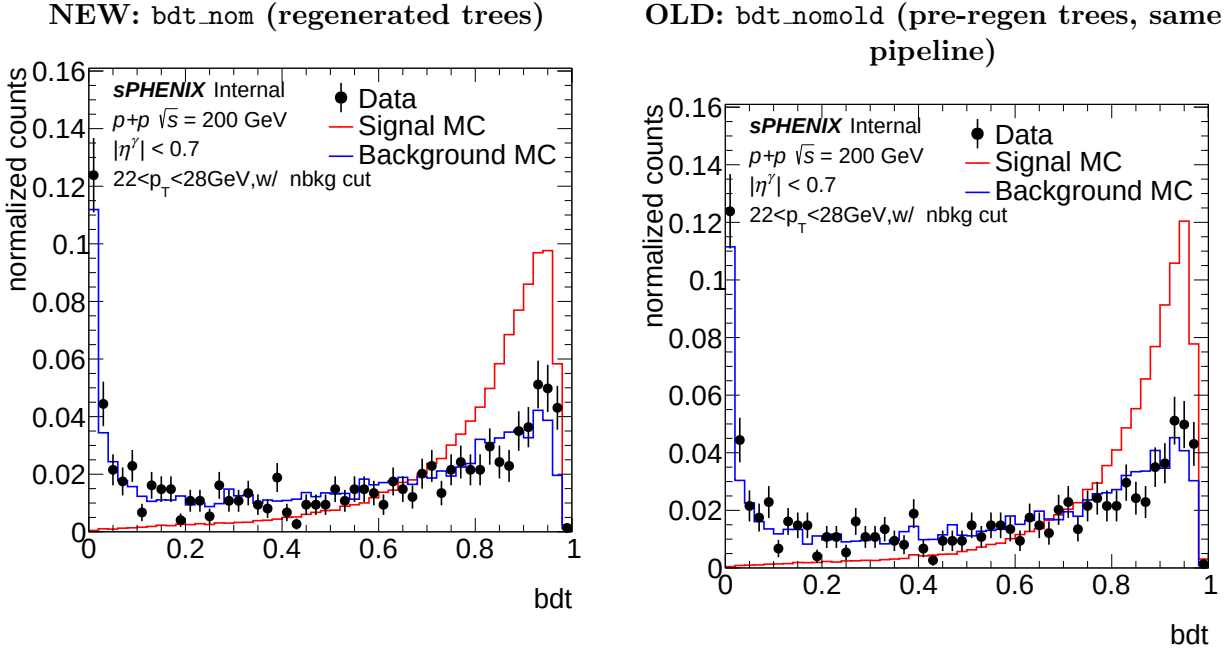


Figure 5: [Item 4 — shower-shape BDT score, **THE DRIVER**.] Signal-MC BDT photon-ID score (1.5 mrad, 22–28 GeV) from `plot_showershapes.C`. The NEW signal BDT shifts to lower values (mean 0.796 vs 0.814; fraction with $\text{BDT} > 0.5$: 0.921 vs 0.935) — the direct cause of the ID-efficiency drop.

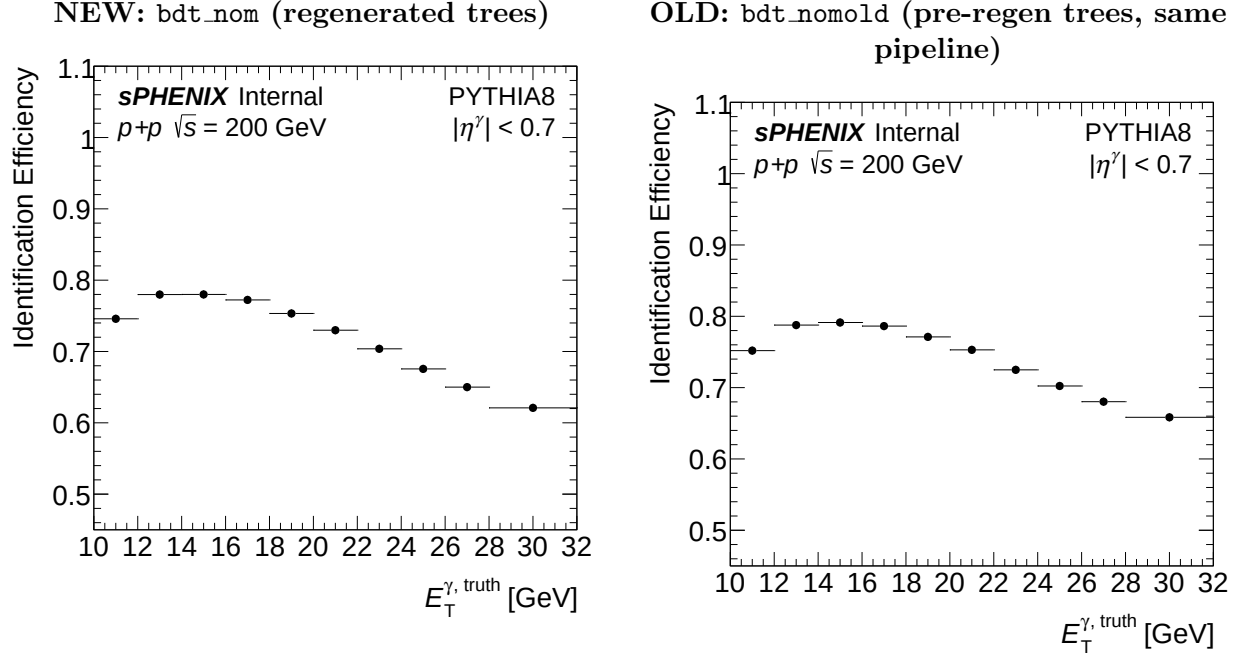


Figure 6: [Item 5 — ID efficiency, the driver.] Photon-ID efficiency from `plot_efficiency.C`. The NEW ID efficiency is lower above ~ 20 GeV, propagating the BDT-score shift of Fig. 5.

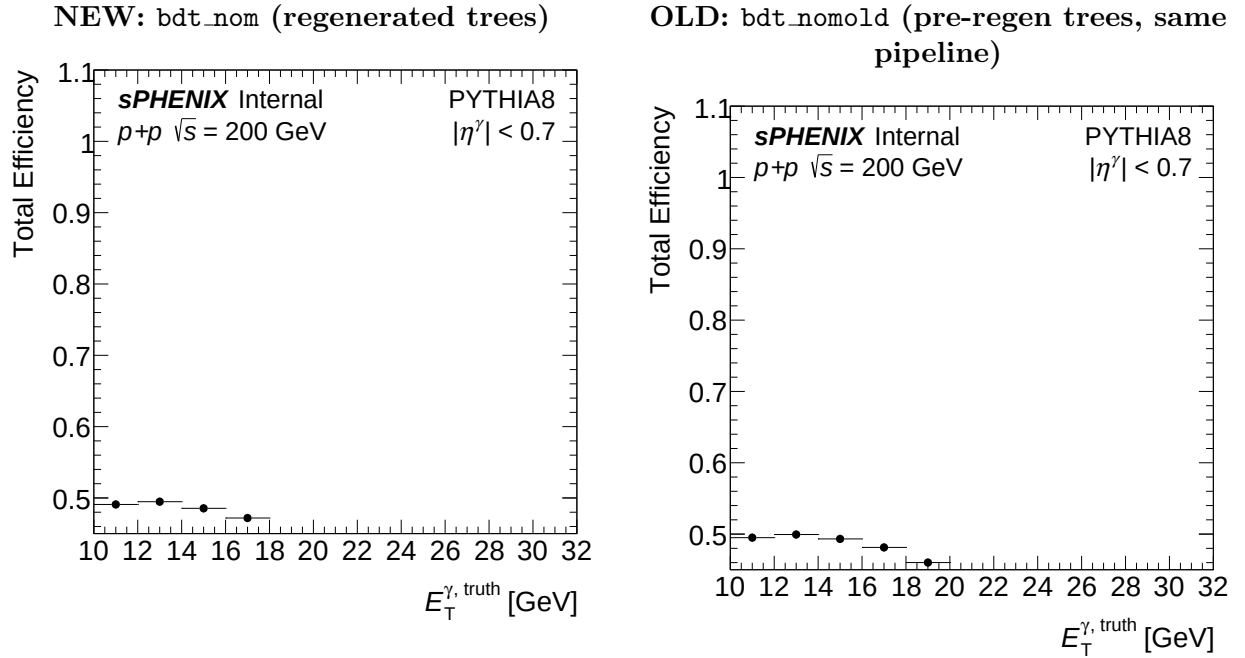


Figure 7: [Item 5b — total efficiency.] Total efficiency from `plot_efficiency.C`. The total efficiency is lower at high E_T (reconstruction and isolation efficiency are unchanged).

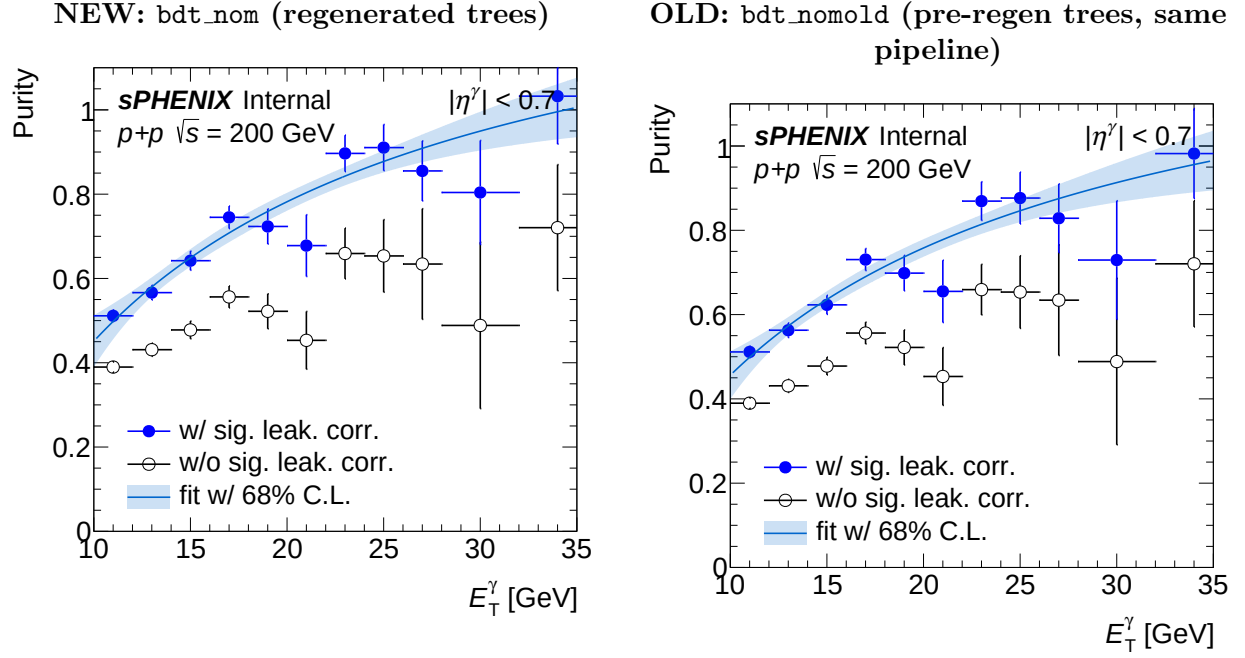


Figure 8: [Item 6 — purity as a function of E_T (incl. signal leakage).] ABCD purity P vs E_T from `plot_paper_purity.C`: filled = with signal-leakage correction (the nominal purity used in the cross section), open = without, blue band = fit with 68% C.L. New vs old: the leakage-corrected purity is essentially unchanged (new/old = 1.000); the signal-leakage correction itself (filled–open gap) is $\sim 20\%$ larger at high E_T for NEW.

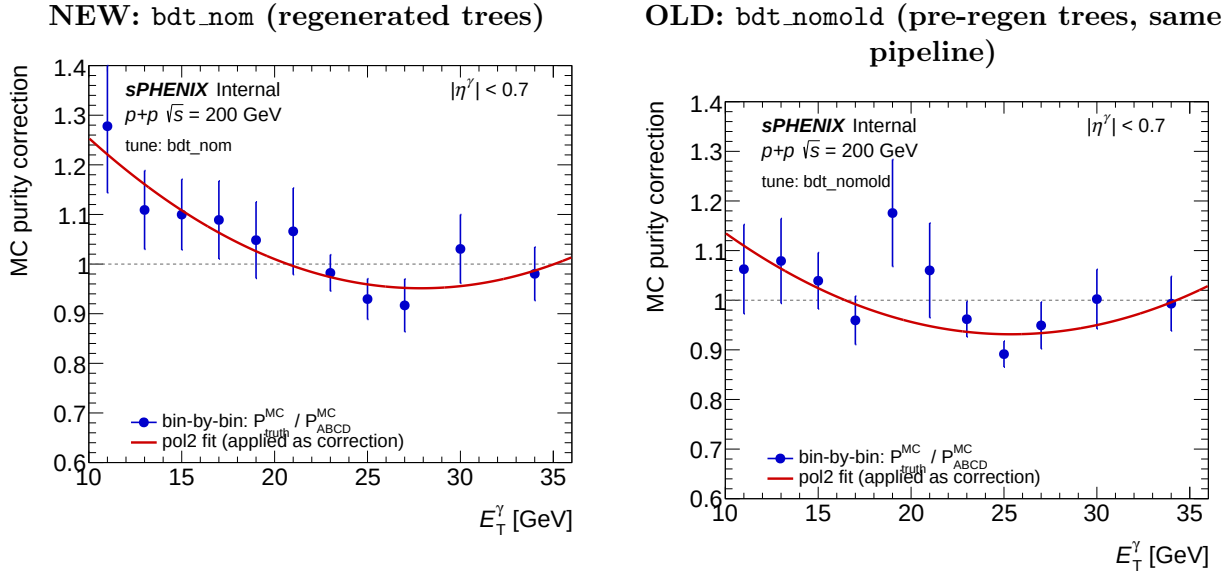


Figure 9: [Item 7 — purity non-closure.] ABCD purity non-closure correction from `plot_mc_purity_correction.C`. The ABCD purity is unchanged (new/old = 1.000).

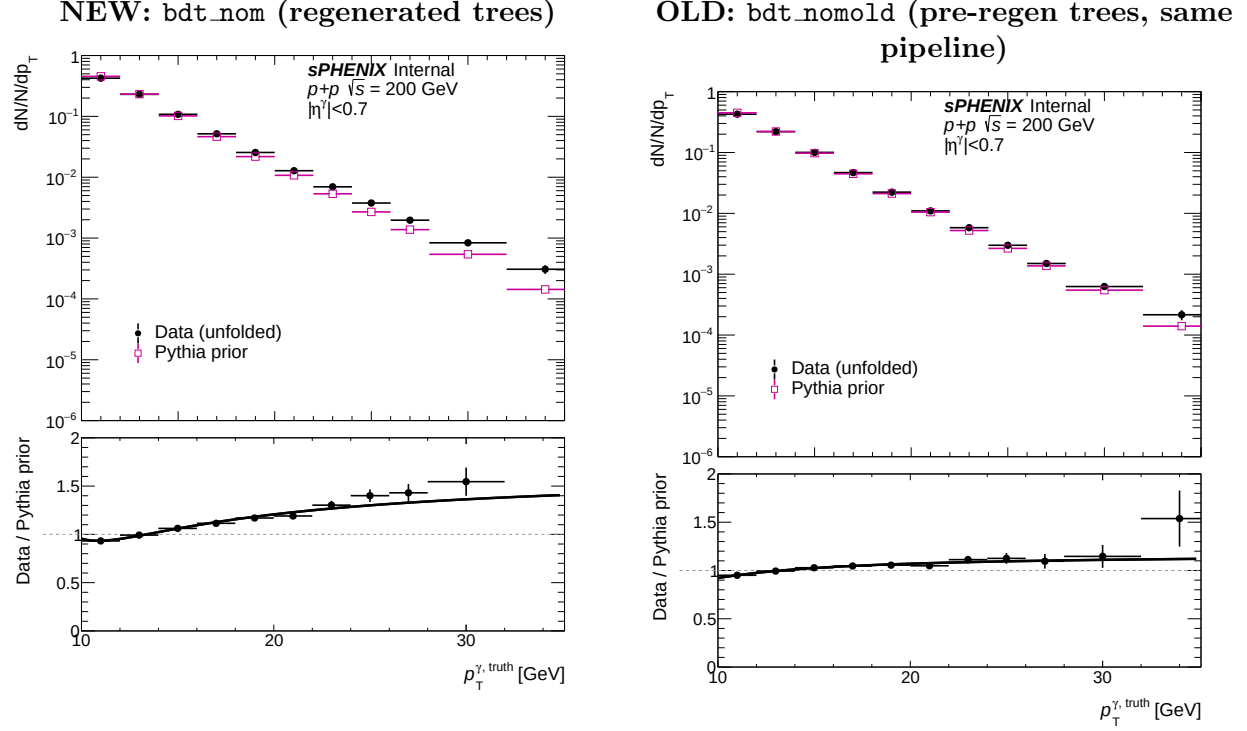


Figure 10: [Item 8 — unfolding prior + reweight.] Unfolding prior and data/prior reweight from `plot_reweight.C`. The NEW prior is softer relative to data, so the reweight factor is larger (mean data/prior 1.21 vs 1.06).

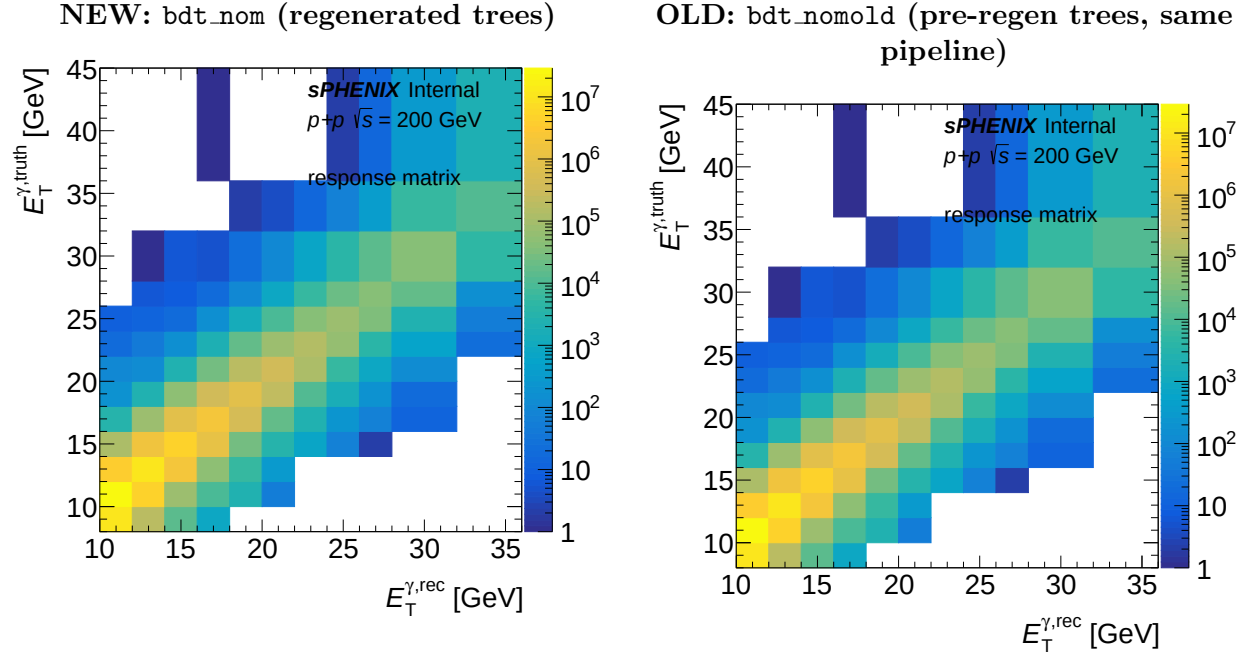


Figure 11: [Item 9 — unfolding response matrix.] Unfolding response matrix from `plot_response.C`. The NEW matrix has broader migration (diagonal fraction 0.203 vs 0.255).

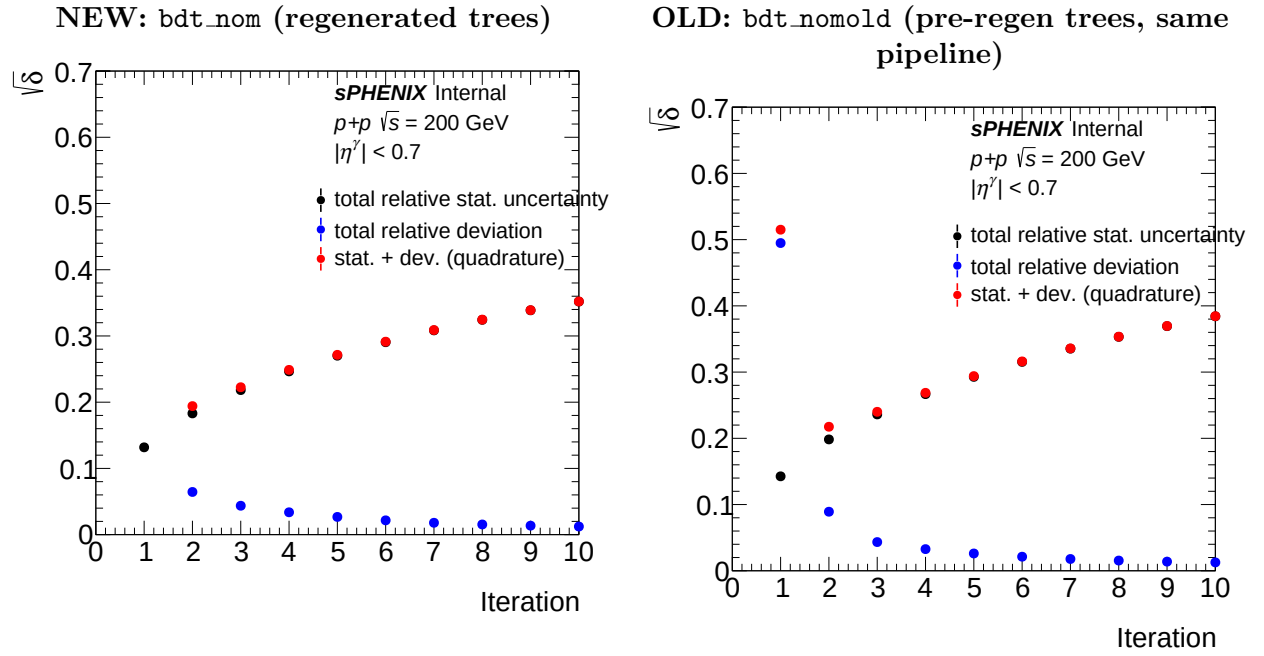


Figure 12: [Item 10 — unfolding closure / iterations.] Iterative-unfolding closure from `plot_unfold_iter.C`. Both are well converged; NEW is slightly faster (iter2/iter1 shape change 0.064 vs 0.089).